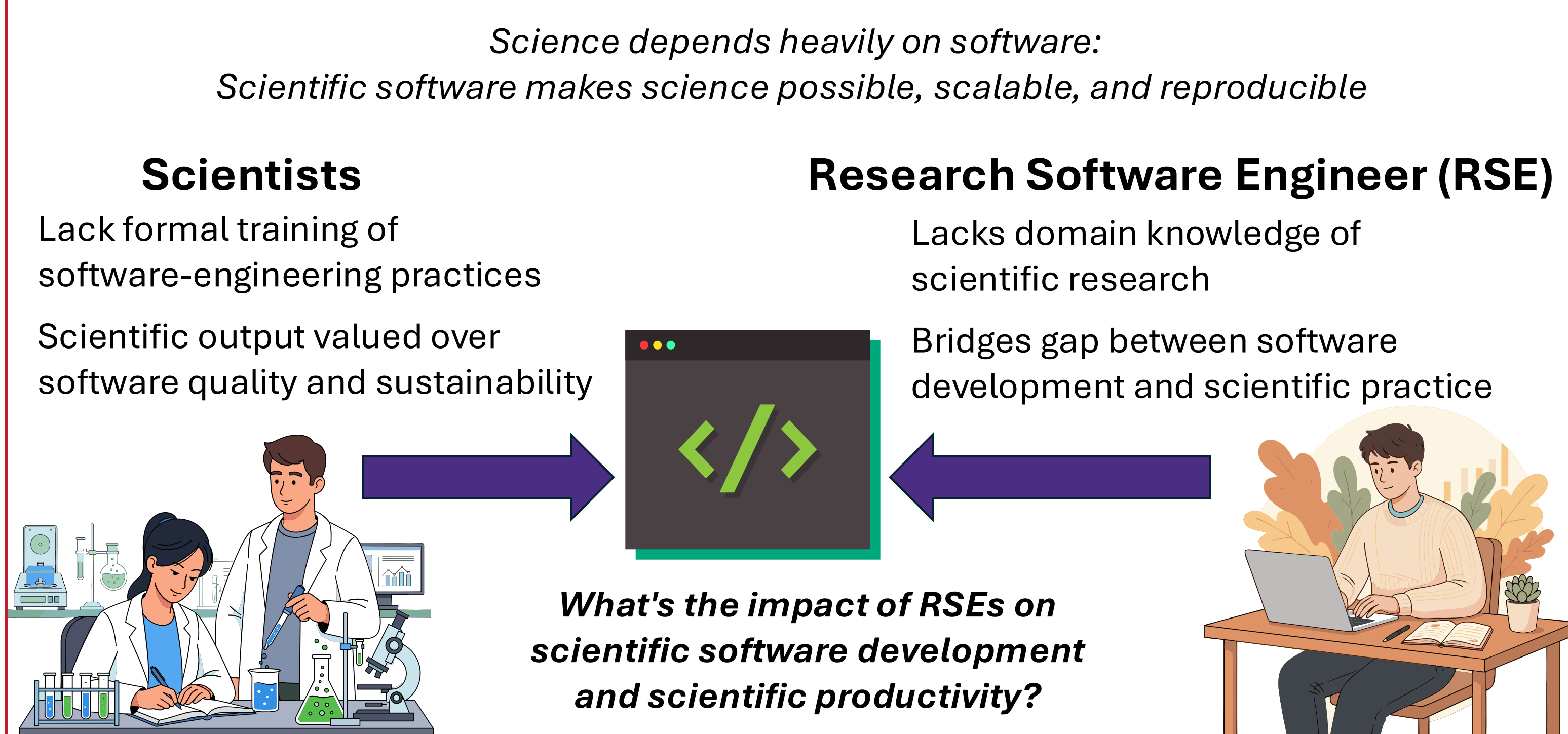


Understanding the Impact of Research Software Engineers on Scientific Software Development: A Mixed-Methods Study

Thomas Bock, Will Sutherland, Curtis Atkisson, Anissa Tanweer, James D. Herbsleb, Bogdan Vasilescu

Motivation



Qualitative Component

Preliminary results from the interviews:

17 interviews so far (8 with RSEs, 9 with RSE managers)

Different understanding of RSE role

Benefits and challenges of long-term vs. short-term engagement

Long-term:
Time to develop domain knowledge

Short-term:
Ability to change projects

Teach scientists software engineering practices

Focus on software engineering work and let scientists do programming

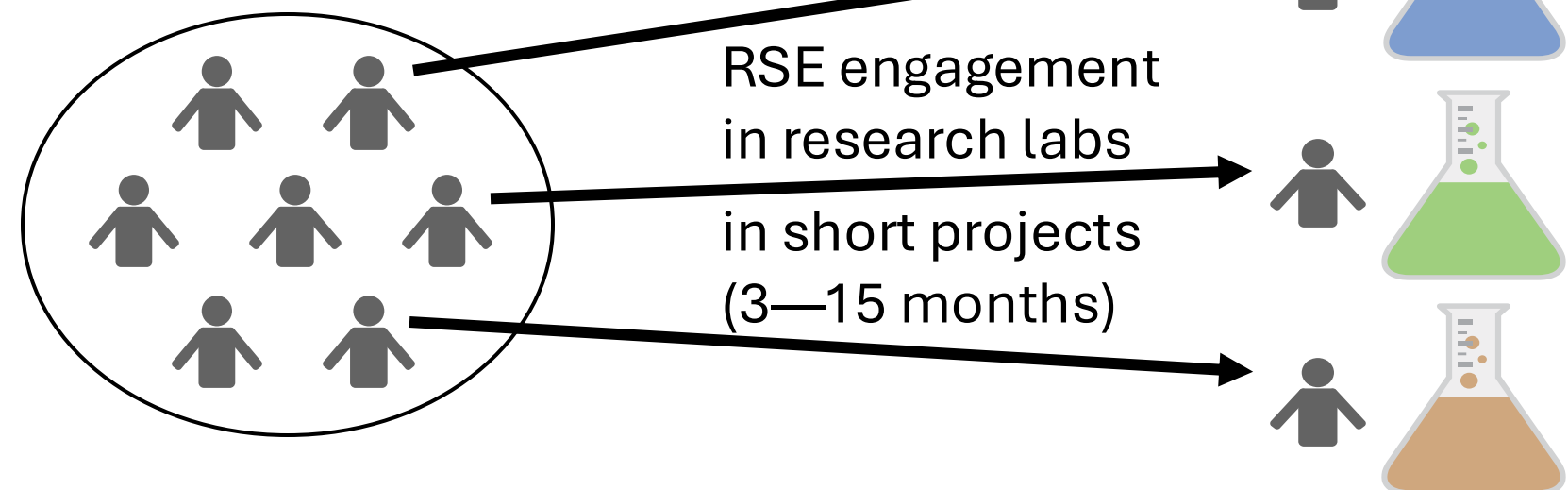
Long-term:
Understand research group's practices

Short-term:
Learn new tools regularly

Next steps: explore these and other contrasts, to understand how they might be systematic or structural

Setting and Approach

Focal group: Centralized RSE teams at large US research universities



Mixed-methods approach:



Qualitative component:

Approximately 80 semi-structured interviews with RSEs, RSE managers, and scientists



Quantitative component:

Repository data analysis, heuristics to identify RSEs, measuring impact on software practices and scientific output

Research Questions:

RQ1: What **challenges** do RSEs and scientists face in effective collaboration?

RQ2: How does RSE collaboration impact the software **development process** and culture?

RQ3: What is the **technical impact** on code quality and productivity?

RQ4: What is the **scientific impact** on scholarly output and software reuse?

Quantitative Component

Preliminary results from repository analysis:

based on 54 scientific projects with RSE involvement

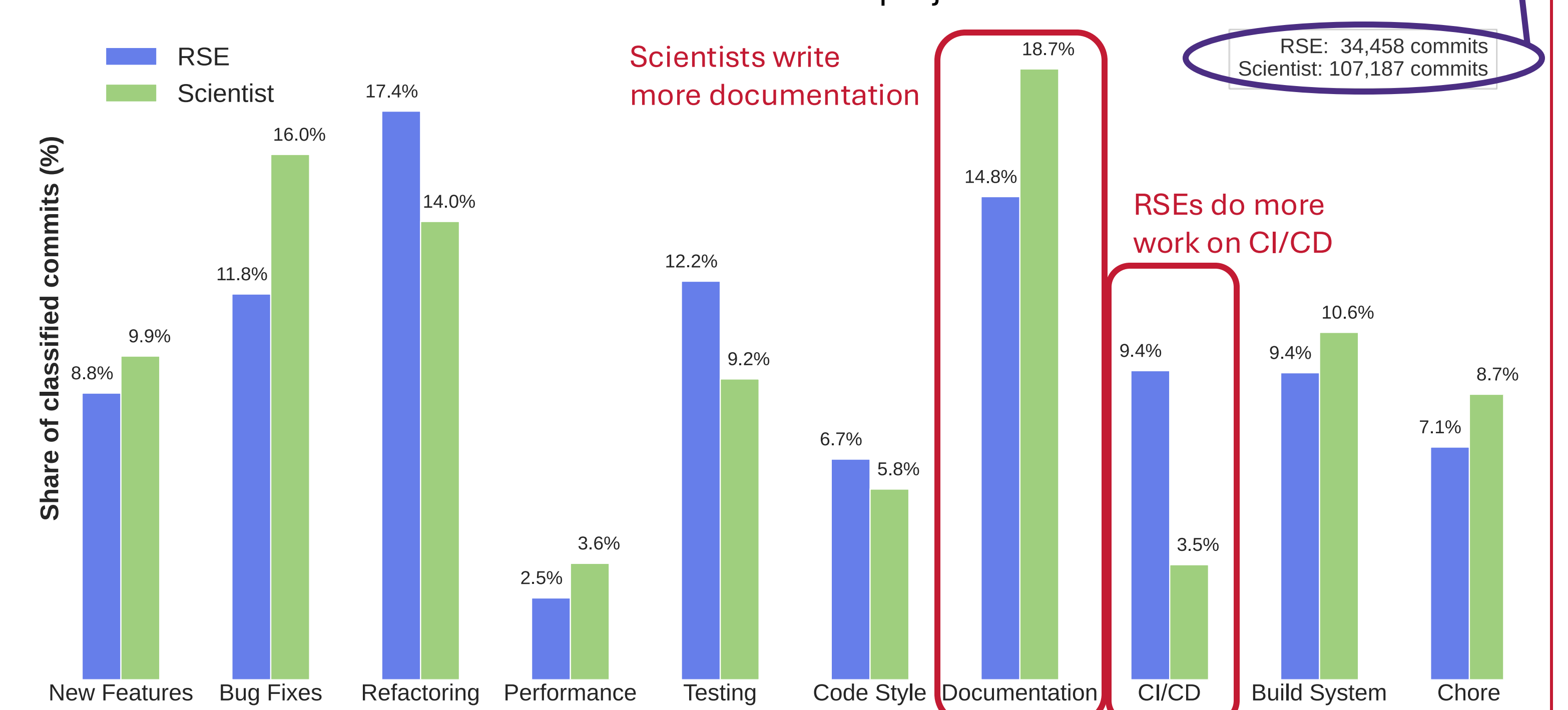
Per-project statistics	Min	Median	Max
Commits	17	2 626	31 035
Files	14	755	10 215
Project time period	0.1 years	2.1 years	18.4 years
RSEs with commits	0	2	5
Scientists with commits	0	4	204

Questions yet to be answered:

• What happened in these **long-lasting projects before the establishment of the RSE role?**

• How many of these non-RSEs are **really scientists?** Are there quasi-RSEs (e.g., students, etc.)?

Commit types of RSEs and scientists using Conventional Commits Specification (CCS):
based on 39 RSEs and 1017 scientists in 54 scientific projects with RSE involvement



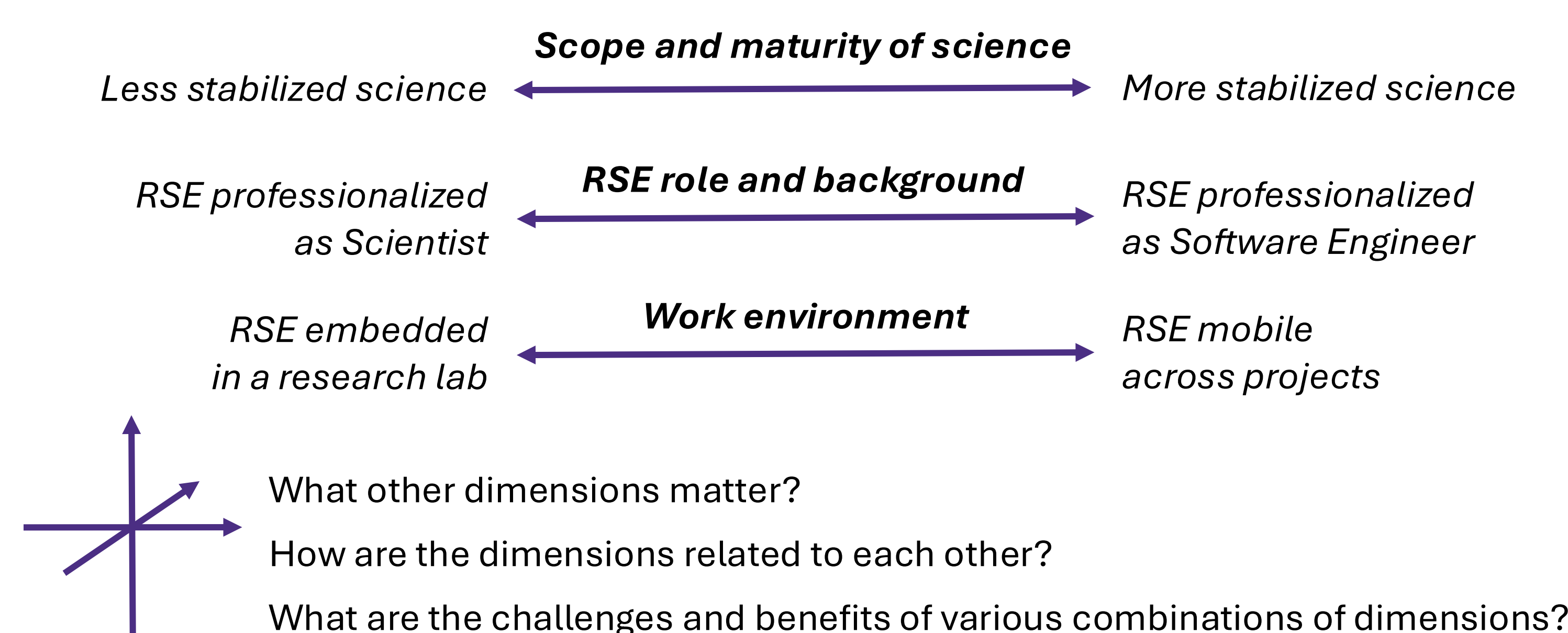
Challenges

Challenges in assessing the impact of RSEs:

- Who is an RSE?
- Do **RSEs' backgrounds** influence the impact?
- Do **scientists' backgrounds** influence the impact?
- Does the **scope of a scientific project** influence the impact?
- Does the **maturity of a scientific project** influence the impact?



Some dimensions of difference in RSE work:



Outlook

How does RSEs' work change with the rise of Generative AI usage?

RSEs' reflections from our interviews:

Scientists come with more sophisticated problems to RSEs as AI can solve the simpler ones now

"you're kind of sticking your head in the sand at this point if you're not at least trying"

RSEs get more AI slop to review from scientists and other RSEs

